

Education and Upward Mobility

Economics of Public and Social Issues

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2026-08-25

Plan for today

- ▶ Correlation of Mobility and Education
- ▶ How to measure how much mobility each college provides

Education

Education and Upward Mobility

- ▶ Education is widely viewed as one of the most scalable pathways to upward mobility
- ▶ But there is growing concern that education no longer provides a strong pathway to opportunity in the U.S.
 - ▶ U.S. students perform worse on standardized tests on average than in many European countries despite higher spending on schools
 - ▶ Sharp differences in quality of schools across districts
 - ▶ Rising costs of college lack of access for low-income students
 - ▶ Concern that some colleges (e.g., for-profit institutions) may not produce good outcomes

Education and Upward mobility

- ▶ How can we improve education in America?
 - ▶ Traditionally, measuring impacts of education systematically was difficult
 - ▶ Administrative data from colleges and school districts are giving us a more scientific understanding of the “education production function”
- ▶ Start with higher education, then turn to K-12 education
Reference: Chetty, Friedman, Saez, Turner, Yagan. “Mobility Report Cards: The Role of Colleges in Intergenerational Mobility” NBER WP 23618 (2017)

How Do Colleges Shape Income Mobility in the U.S.?

- ▶ How does the higher education system affect intergenerational income mobility in the U.S.?
 - ▶ In principle, higher education can provide a pathway to upward mobility that is not directly shaped by the neighborhood where a child happens to grow up
 - ▶ But if children from higher-income families tend to attend better colleges, higher education system may not promote mobility
 - ▶ Colleges could actually increase intergenerational persistence of income if disparities in college attendance are sufficiently large

Effect of Higher Education System on Mobility

- ▶ Effect of higher education system on mobility depends upon three factors:
 - ▶ [Inputs] Parental income distributions by college
 - ▶ [Outputs] Students' earnings outcomes conditional on parental income by college
 - ▶ [Causal share] Portion of variation in students' earnings outcomes that is due to colleges' causal effects

Estimating the Three Parameters: Data

- ▶ Chetty et al. (2017) estimate these three parameters using data covering all college students in the U.S. from 1999-2013 (30 million students)
- ▶ Combine information from three sources to construct an anonymized dataset:
 - ▶ Parental and Student Income from income tax records
 - ▶ College attendance from 1098-T tax data and Pell grant data
 - ▶ SAT scores from College Board

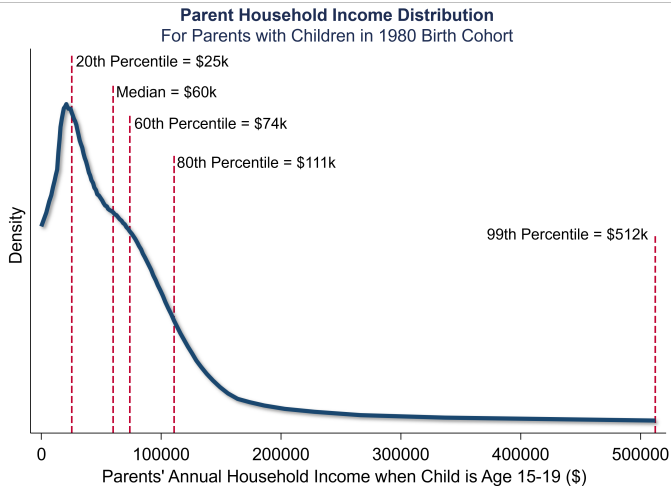
Note: all statistics are based on college attendance (not completion)

Parents' Income Distributions by College: Income Segregation in the American Higher Education System

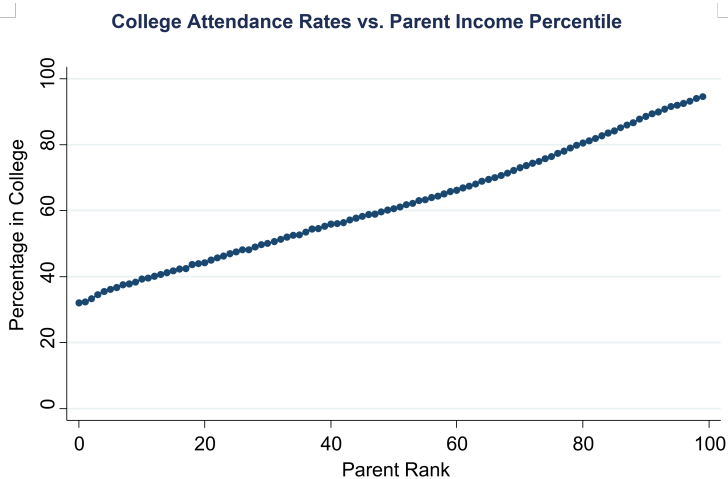
Measuring Parents' Incomes

- ▶ Parent income: average pre-tax household income during five year period when child is aged 15-19
- ▶ Focus on percentile ranks, ranking parents relative to other parents with children in same birth cohort

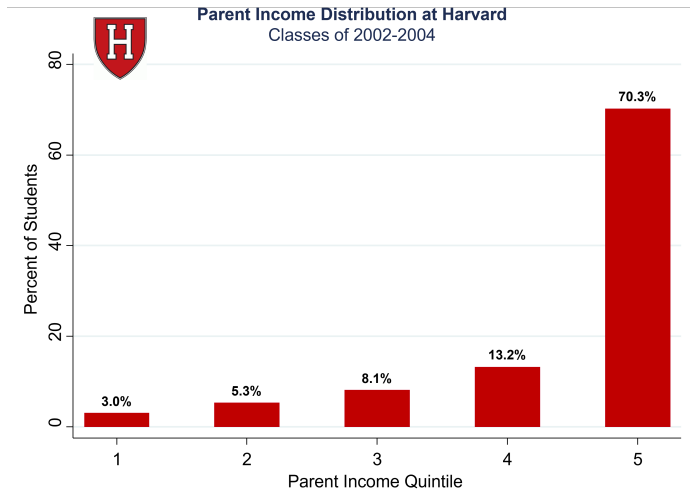
Parent Household Income Distribution



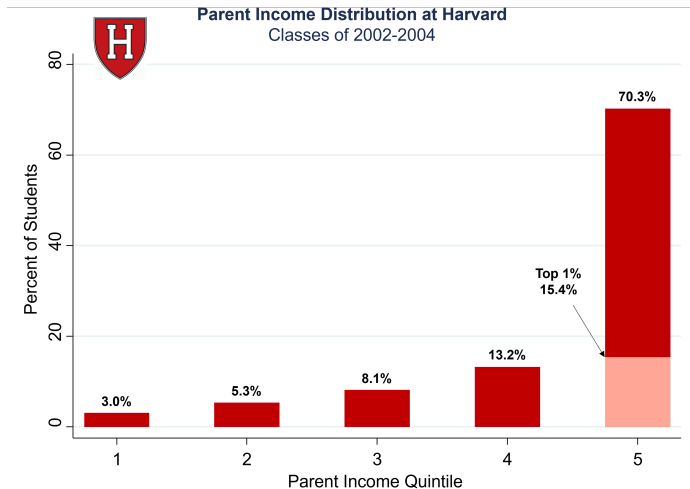
College Attendance Rates vs. Parent Income Percentile



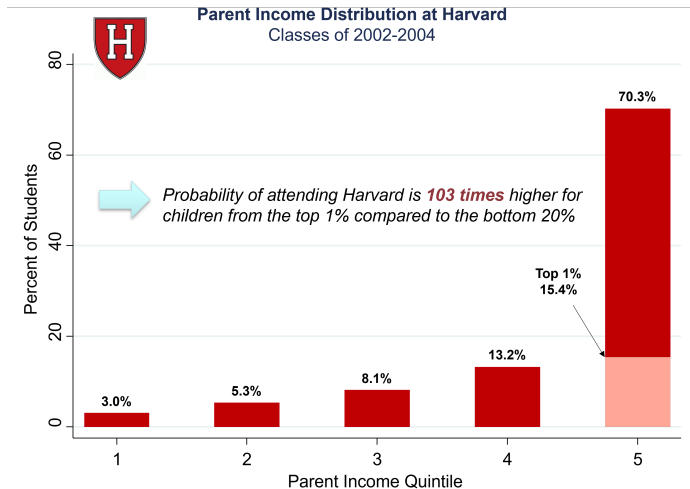
Parent Income Distribution at Harvard



Parent Income Distribution at Harvard

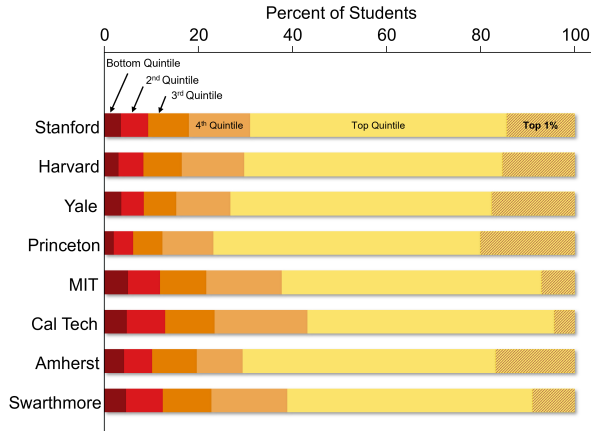


Parent Income Distribution at Harvard

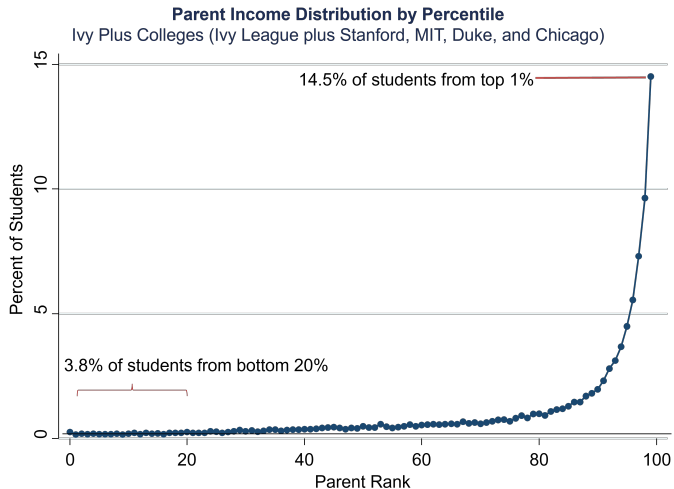


Parental Income Distribution of Students at Some Highly Selective Colleges

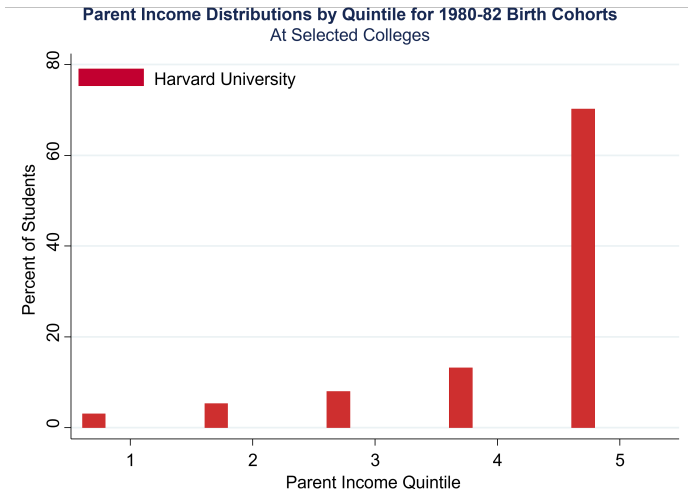
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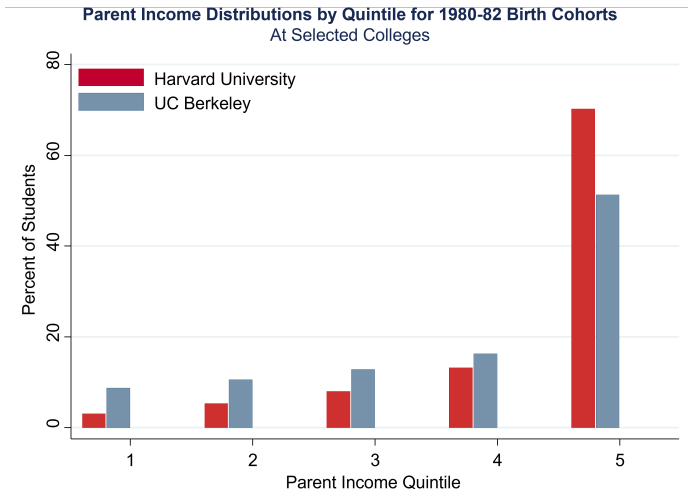
Income Composition of Ivy Leagues



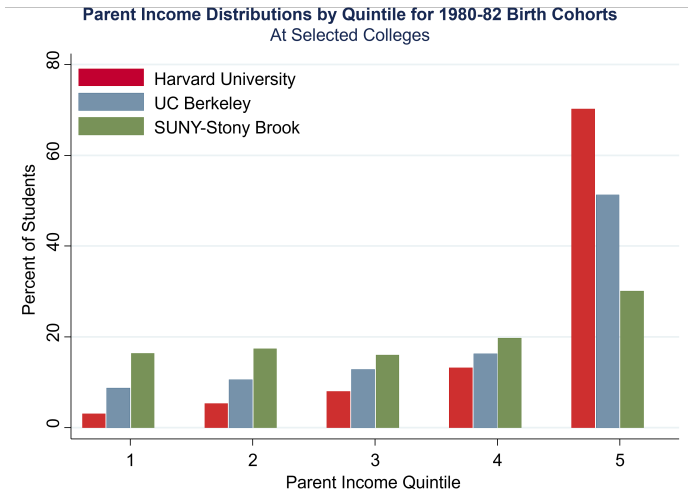
Parent Income Distributions by Quintile for 1980-82 Birth Cohorts - Selected Colleges



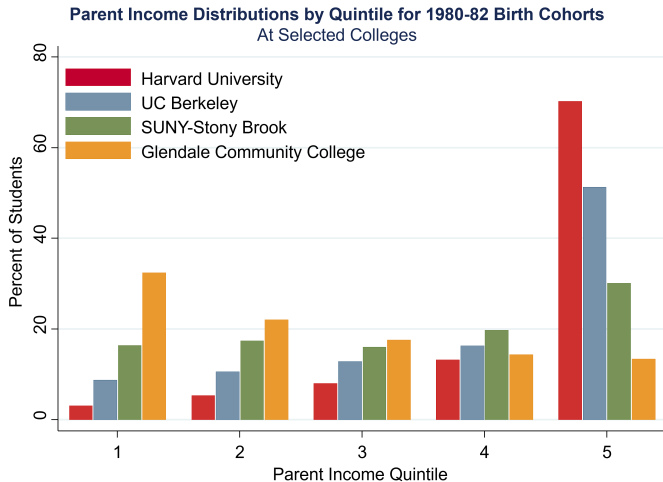
Parent Income Distributions by Quintile for 1980-82 Birth Cohorts - Selected Colleges



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Parent Income Distributions by Quintile for 1980-82 Birth Cohorts - Selected Colleges



How does UTSA look?

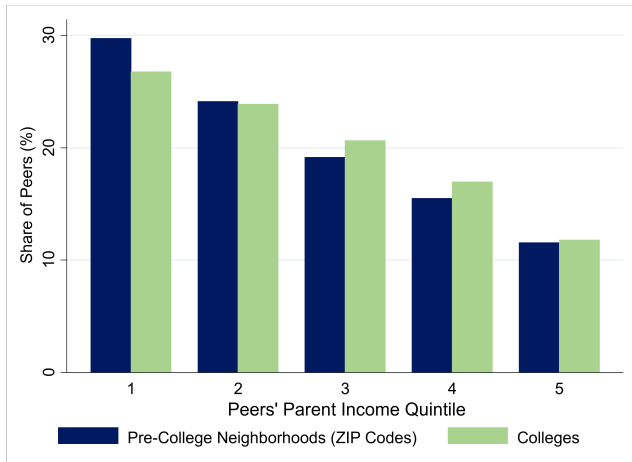
- ▶ All Schools: <https://www.nytimes.com/interactive/projects/college-mobility>
- ▶ UTSA: <https://www.nytimes.com/interactive/projects/college-mobility/university-of-texas-at-san-antonio>

Parental Income Segregation Across Colleges

- ▶ Sharp differences in parental income distributions across colleges there is significant segregation across colleges
- ▶ Useful benchmark to quantify magnitude: compare to degree of segregation across neighborhoods
- ▶ Common perception: colleges foster greater interaction between children from different socioeconomic backgrounds than places in which they grew up

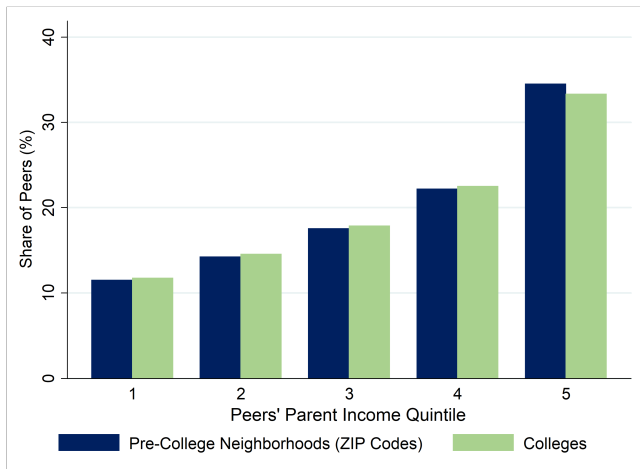
Parental Income Distribution of Peers of Children from Bottom Quintile

Parental Income Distribution of Peers of Children from Bottom Quintile



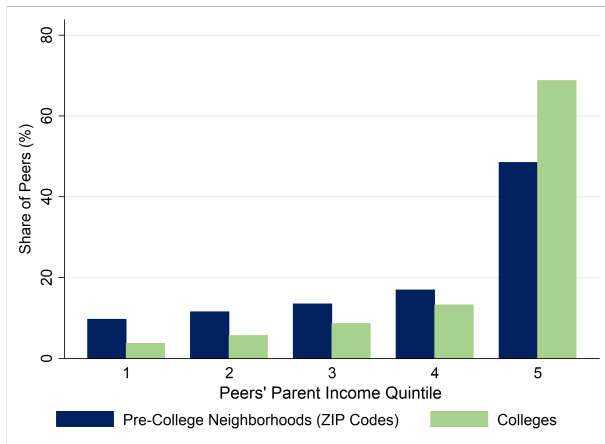
Parental Income Distribution of Peers of Children from Top Quintile

Parental Income Distribution of Peers of Children from Top Quintile



Parental Income Distribution of Peers of Ivy-Plus College Students from Top Quintile

Parental Income Distribution of Peers of Ivy-Plus College Students from Top Quintile

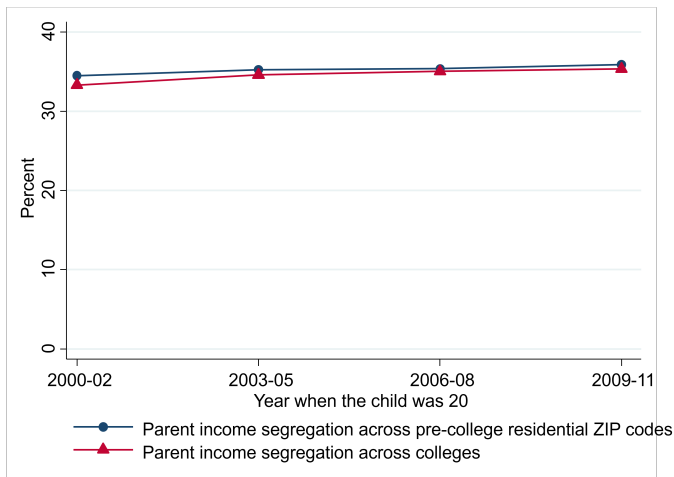


Trends in Income Segregation

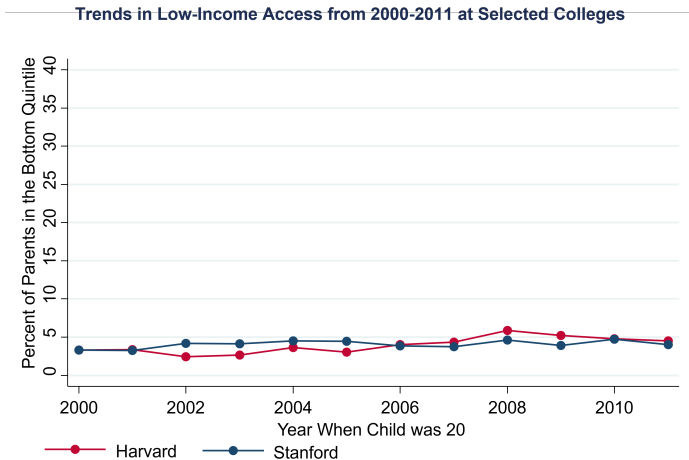
- ▶ Preceding estimates are based on children born between 1980-82, who attended college in the early 2000s
- ▶ Substantial changes in higher education system since that time, e.g. substantial changes in financial aid and tuition policies
- ▶ How has income segregation across colleges changed in recent years?

Fraction of Peers from the Top Quintile for Children from the Top Quintile

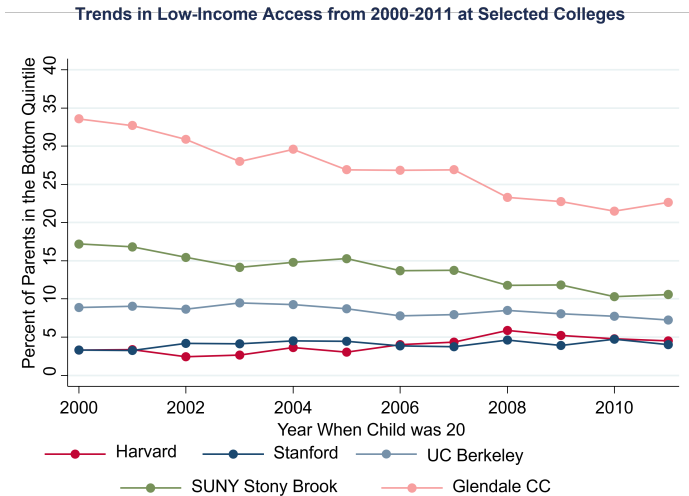
Fraction of Peers from the Top Quintile for Children from the Top Quintile



Trends in Low-Income Access from 2000-2011 at Selected Colleges



Trends in Low-Income Access from 2000-2011 at Selected Colleges



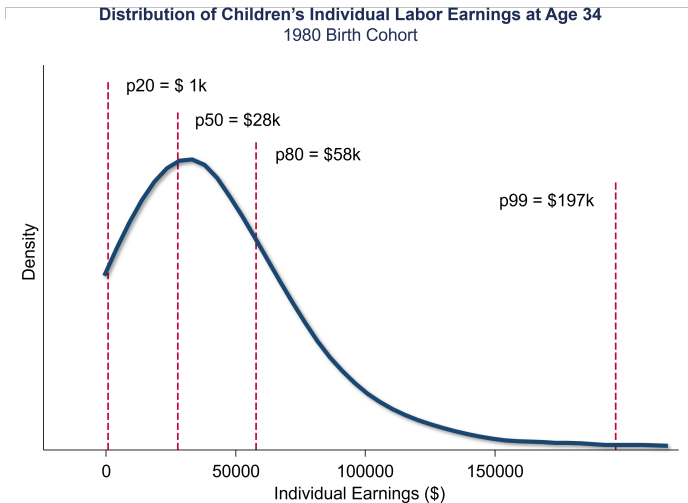
Outcomes: Students' Earnings Distributions

Students' Earnings Outcomes

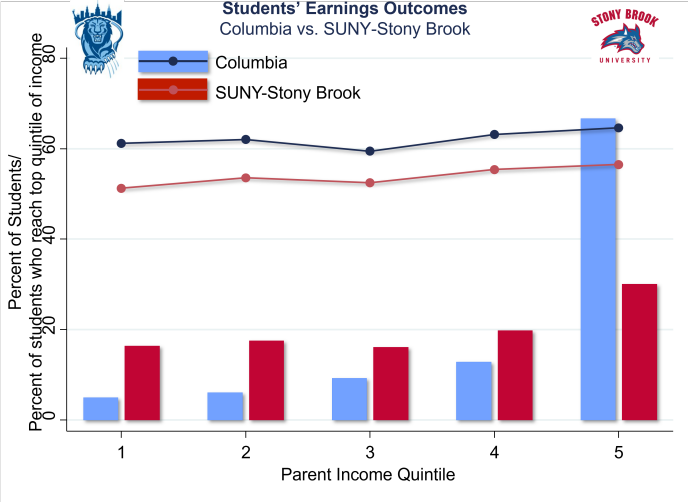
- ▶ Measure children's individual earnings in their mid-30s
- ▶ Define percentile ranks by ranking children relative to others in same birth cohort

Distribution of Children's Individual Labor Earnings at Age 34

1980 Cohort



Students' Earnings Outcomes: Columbia vs. SUNY-Stony Brook



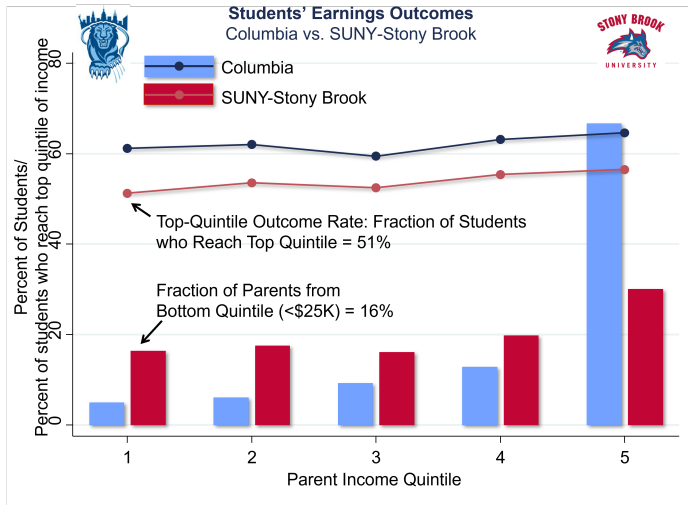
Students' Earnings Outcomes

- ▶ Key lesson: most of the gap in outcomes between children from low vs. high-income families is explained by differences between rather than within colleges
- ▶ Raises possibility that reallocating student across colleges could potentially have a significant impact on intergenerational mobility
 - ▶ If gap in outcomes by parental income were large even within a given college, there would be little scope to have an impact through changes in college admissions policies

Differences in Mobility Rates Across Colleges

- ▶ We can combine data on parents' incomes and students' outcomes to characterize colleges' mobility rates
 - ▶ At which colleges in America do the largest number of children come from poor families and end up in the upper middle class?

Students' Earnings Outcomes: Columbia vs. SUNY-Stony Brook



Measuring Mobility Rates

Define a college's mobility rate (MR) as the fraction of its students who come from bottom quintile and end up in top quintile

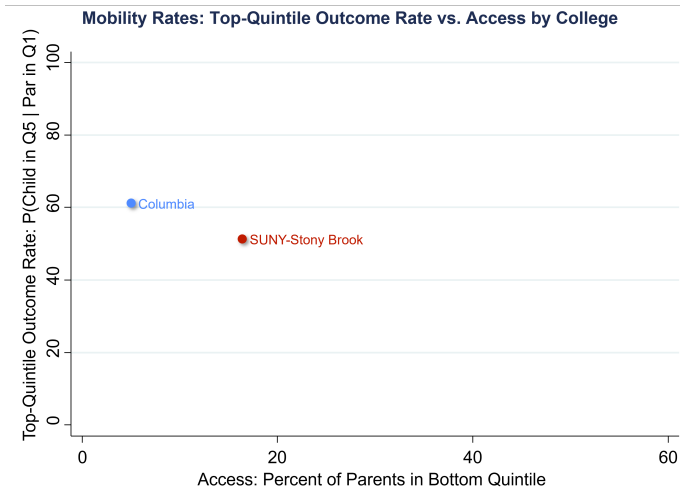
Mobility rate is:

$$\underbrace{\text{Mobility Rate}}_{\text{Frac. of Parents in Q1 and Children in Q5}} = \underbrace{\text{Low-Inc. Access}}_{\text{Frac. of Parents in Q1}} \times \underbrace{\text{Top-Quintile Rate}}_{\text{Frac. of Students who Reach Q5 Given Parents in Q1}}$$

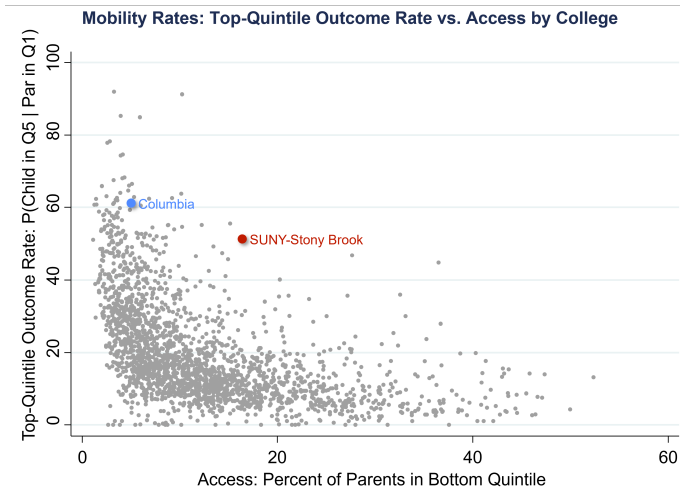
At SUNY:

$$8.4\% = 16\% \times 51\%$$

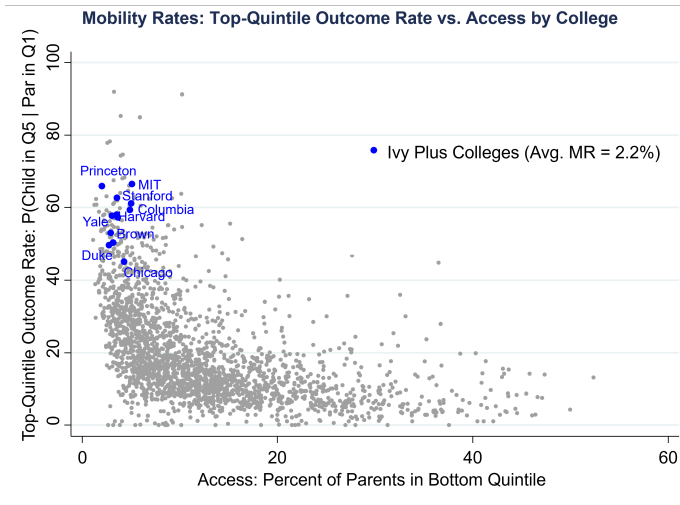
Mobility Rates: Top-Quintile Outcome Rate vs. Access by College



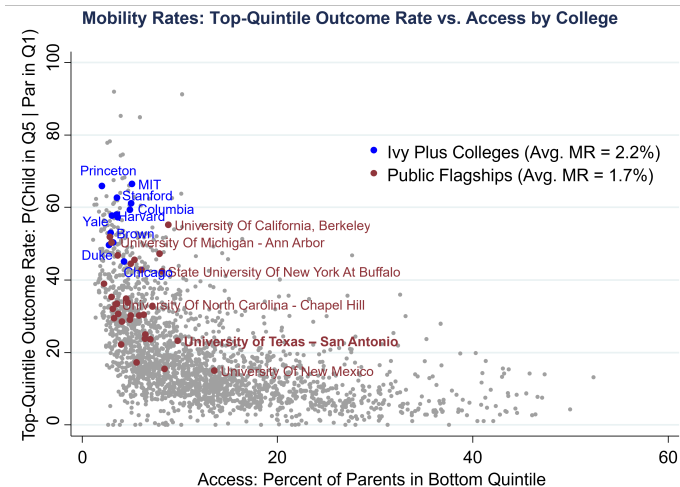
Mobility Rates: Top-Quintile Outcome Rate vs. Access by College



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Mobility Rates: Top-Quintile Outcome Rate vs. Access by College



Top 10 Colleges in America By Bottom-to-Top Quintile Mobility Rate

Fraction of Students who come from Bottom Fifth and End up in Top Fifth

Top 10 Colleges in America By Bottom-to-Top Quintile Mobility Rate

Fraction of Students who come from Bottom Fifth and End up in Top Fifth

